

1. Introduction

The PolarFire® SoC FPGA Icicle Kit (MPFS250T-FCVG484E) is a development platform from Microchip designed for evaluating the PolarFire® SoC FPGA, which combines a hardened RISC-V multi-core processor subsystem with low-power FPGA fabric. It enables developers to explore both Linux® - based applications and bare-metal Firmware on a flexible and open-source RISC-V architecture.

This guide helps you to power up the board, connect to the on-board serial interfaces and boot into Linux.

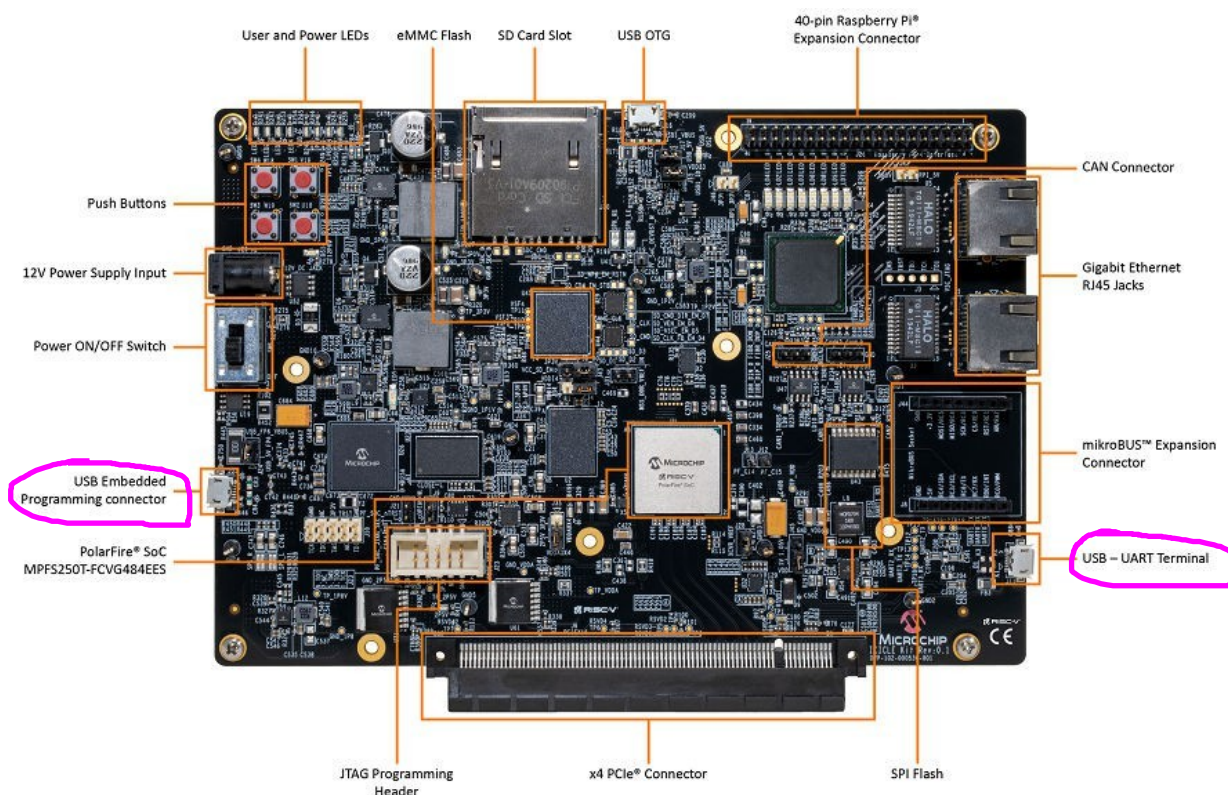
The following table lists the contents of the PolarFire SoC Icicle Kit.

Table 1-1. Kit Contents

Item	Quantity
PolarFire® SoC FPGA Icicle Kit featuring the MPFS250T-FCVG484E device with 254K logic elements	1
12V/5A wall-mounted power adapter	1
Ethernet cable	1
USB 2.0 micro AB connector for UART interface to PC	1
QuickStart card	1

The following figure shows the top view of PolarFire SoC Icicle Kit.

Figure 1-1. PolarFire® SoC Icicle Kit—Top View



1.1. Jumper Settings

The following table lists the factory default settings for the Icicle Kit.

Table 1-2. Icicle Kit Setting

Jumper	Default Settings
J15	Open
J17	Open
J24	Open
J28	Closed
J31	Open
J34	1 and 2
J35	1 and 2
J43	1 and 2
J45	1 and 2
J46	Closed

The following table lists the default settings of FlashPro jumpers.

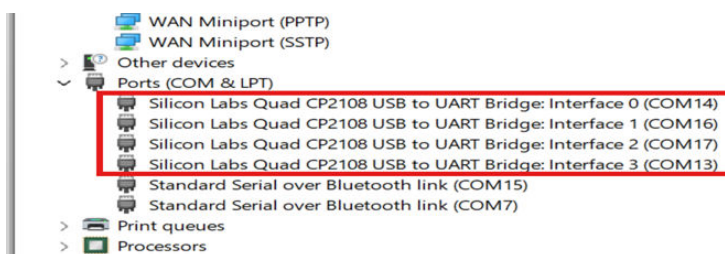
Table 1-3. FlashPro Jumpers

Jumper	Default Settings
J9	Select between <u>embedded FlashPro</u> and <u>external FlashPro</u> <u>J9 Closed: On-board embedded FlashPro connected to the J33 USB Port</u> <u>J9 Open: External FlashPro connected to the J23 header</u>
J21	JTAG nTRST interface pull down enable (leave open)

1.2. Running the Kit

To run the kit, perform the following steps:

1. To run the kit, ensure the jumper settings are correct.
2. Connect the Icicle Kit's USB to UART terminal connector (J11) to your host computer.
3. Plug in the 12V power supply to the Icicle Kit. To switch On, turn the power on/off.
4. Perform the following steps to determine the correct COM ports (Windows) or TTY devices (Linux):
 - On Windows[®] host:
 - a. Open Device Manager: On the keyboard, press Windows+X and click **Device Manager**.
 - b. Expand the ports (COM and LPT) section. The following figure shows the interface, if your USB-to-UART device is recognized.

Figure 1-2. Device Manger

Note: If your computer does not recognize the COM port, install the required drivers from [Silicon Labs USB to UART Bridge VCP Drivers](#).

- c. Open a serial terminal:
 - i. Use a serial terminal application such as PuTTY or TeraTerm.

- ii. Open serial connections to the four UART interfaces available through the USB-UART port. The following table lists the settings to configure the serial port.

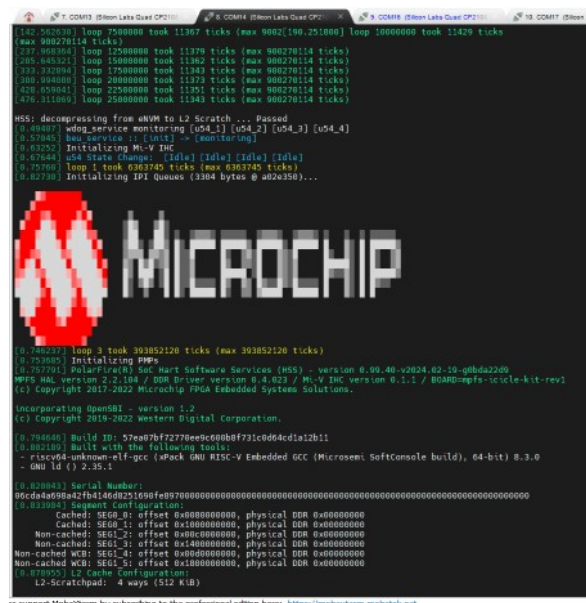
Table 1-4. Serial Port Settings

Parameter	Value
Baud rate	115200
Data bits	8
Parity	None
Stop bits	1
Flow Control	None

- d. Identify the correct COM ports:
- Silicon Labs Quad CP2108 USB to UART Bridge: Interface 0
 - Silicon Labs Quad CP2108 USB to UART Bridge: Interface 1
- On Linux host:
- Use a serial terminal application such as Minicom Screen or Picocom.
 - Open a terminal and run the following command:


```
mesg | grep tty
```
 - Look for entries such as `/dev/ttyUSB0`, `/dev/ttyUSB1`.
 - To open `/dev/ttyUSB0` using minicom with a common baud rate (for example, 115200), run:


```
sudo minicom - D /dev/ttyUSB0 -b 115200
```
5. Turn the power switch off, then on again. You should now see boot messages in your terminal windows. UART Output:
- Interface 0: Displays Hart Software Services boot messages.

Figure 1-3. Interface 0

- Interface 1: Displays U-Boot and Linux boot messages and provides a Linux shell prompt.

Figure 1-4. Interface 1

```

7. COM13 (Silicon Labs Quad CP) 8. COM14 (Silicon Labs Quad CP)
DDR training ... Passed ( 2940 ms)
[3.35216] DDR-Lo size is 32 MiB
[3.39799] DDR-Hi size is 1888 MiB

OpenSBI v1.2

OpenSBI

Platform Name      : Microchip PolarFire(R) SoC
Platform Features  : medeleg
Platform HART Count : 5
Platform IPI Device : aclint-mswi
Platform Timer Device : aclint-mtimer @ 1000000Hz
Platform Console Device : mmuart
Platform HSM Device : mpfs_hsm
Platform PMU Device : ---
Platform Reboot Device : mpfs_reset
Platform Shutdown Device : mpfs_reset
Firmware Base      : 0xa0000000
Firmware Size      : 149 KB
Runtime SBI Version : 1.0

Domain0 Name       : root
Domain0 Boot HART   : 1
Domain0 HARTs       : 1,2,3,4
Domain0 Region00    : 0x000000002000000-0x00000000
Domain0 Region01    : 0x000000002000000-0x00000000

```

6. When the Linux prompt appears, log in using:
 - a. Username: root
 - b. Password: No password needed—just press Enter key.

The Icicle Kit is now running Linux. You are ready to interact with the system.

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